

Balance Tracker (1)

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

You have a bank account with an initial balance.

You will be given several operations. Each operation is either a **deposit** or a **withdraw**.

For a **deposit** operation, the given amount is **added** to the balance.

For a **withdraw** operation, the amount is **subtracted** from the balance only if the current balance is sufficient. Otherwise, the operation is **ignored**.

After processing all operations:

- Print **PROFIT** if the final balance is **greater** than the initial balance.
- Print **LOSS** if the final balance is **less** than the initial balance.
- Print **EVEN** if the final balance is **equal** to the initial balance.

Input

The first line contains two integers b and q ($0 \leq b \leq 10^9$, $1 \leq q \leq 10^5$), representing the initial balance and the number of operations.

Each of the next q lines contains a string op and an integer x ($1 \leq x \leq 10^9$).

The string op is either **deposit** or **withdraw**.

Output

Print one of the following words:

- **PROFIT**
- **LOSS**
- **EVEN**

According to the final balance after processing all operations.

Examples

standard input	standard output
100 4 deposit 50 withdraw 20 deposit 30 withdraw 10	PROFIT
1000 5 withdraw 300 withdraw 200 deposit 100 withdraw 400 withdraw 300	LOSS
500 3 deposit 200 withdraw 100 withdraw 100	EVEN